**CS 215: Task 09: Due: 06-Nov; time as announced on Piazza.**

- Your answers should be in your handwriting. Try to be neat and clear. If an LLM can read your handwriting, we might be able to.
- If you used any source (person or thing) explicitly state it. This is an individual task.
- For each of the two computer-based tasks, add to the zip package `a.pdf`, `a.py`, `b.pdf` and `b.py` as your solution(s).

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1. Suppose the response follows the simple linear regression model

$$Y = \alpha + \beta x + e,$$

where $x$ is fixed, $\mathbb{E}[e] = 0$, and $0 < \beta < 1$. Show that if

$$x < \frac{\alpha}{1-\beta},$$

then

$$x < \mathbb{E}[Y] < \frac{\alpha}{1-\beta}.$$

2. Consider the simple linear regression model

$$Y_i = \alpha + \beta x_i + e_i, \qquad i = 1, \ldots, n,$$

where the predictors are fixed and the errors are independent random variables with mean 0 and variance $\sigma^2 > 0$. Let $A$ and $B$ be the ordinary least squares estimators, and define

$$\widehat{Y}_i = A + Bx_i, \qquad R_i = Y_i - \widehat{Y}_i.$$

Let, as usual, $\bar{x} = n^{-1} \sum_i x_i$ and $S_{xx} = \sum_i (x_i - \bar{x})^2 > 0$.

(a) Derive $\mathrm{Cov}(A, B)$ in terms $\bar{x}$, $\sigma^2$ and $S_{xx}$. Recall that $\mathrm{Var}(B) = \frac{\sigma^2}{S_{xx}}$ as mentioned in class.

(b) Under what condition on the predictor values is $\mathrm{Cov}(A, B) = 0$?

(c) A student writes on Piazza: "The observation $Y_i$ has variance $\sigma^2$. Subtracting the estimated fitted value $\widehat{Y}_i$ adds uncertainty" Dissect the student's argument explaining where the student is right. Is the intended conclusion "$R_i$ must have variance at least $\sigma^2$" true or false?
Also explain intuitively your answer.

3. Suppose the observations follow the simple linear regression model

$$Y_i = \alpha + \beta x_i + e_i, \qquad i = 1, \ldots, n,$$

where the predictors are fixed, $S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2 > 0$, and the errors are independent $N(0, \sigma^2)$ random variables. Consider the alternative parameterization

$$V_i = \gamma + \beta(x_i - \bar{x}) + e_i, \qquad \gamma = \alpha + \beta\bar{x}. \tag{1}$$

Let $(A, B)$ and $(C, B^*)$ denote the maximum likelihood estimators under the original and alternative parameterizations, respectively. Here $C$ estimates $\gamma$, and $B^*$ estimates $\beta$ in Eq. 1. The coefficient estimators maximize the Gaussian likelihood with respect to the coefficients for each fixed $\sigma^2 > 0$.

(a) Reason the equivalence of $V_i$ and $Y_i$ and thus $B^* = B$. Also mathematically show this.

(b) Show that $C = \bar{Y}$ and derive its distribution.

4. The response random variables $Y_1, \ldots, Y_n$ follow the model

$$Y_i = \theta x_i^2 + e_i, \qquad i = 1, \ldots, n,$$

where $x_1, \ldots, x_n$ are fixed known constants, not all zero, and $e_1, \ldots, e_n$ are independent $N(0, \sigma^2)$ random variables, with known $\sigma^2 > 0$. The parameter $\theta \in \mathbb{R}$ is unknown.

(a) Find the least squares estimator of $\theta$.

(b) Derive the maximum likelihood estimator of $\theta$.

(c) Which estimator from that obtained in (a) and (b) is unbiased? Find its variance.

5. This is a computational exercise. Use Python, the numpy and matplot packages. We estimate about 10-20 lines of python code.

   Ten experiments on wood samples result in the following data. The input $x_i$ is specific gravity and $y_i$ is the observed maximum crushing strength parallel to the grain, measured in psi.

| $i$ | $x_i$ | $y_i$ | $i$ | $x_i$ | $y_i$ |
|---|---|---|---|---|---|
| 1 | .41 | 1850 | 6 | .39 | 1760 |
| 2 | .46 | 2620 | 7 | .41 | 2500 |
| 3 | .44 | 2340 | 8 | .44 | 2750 |
| 4 | .47 | 2690 | 9 | .43 | 2730 |
| 5 | .42 | 2160 | 10 | .44 | 3120 |

   Consider $Y_i = \alpha + \beta x_i + e_i$, where the inputs are fixed and the errors are independent $N(0, \sigma^2)$ random variables. We write $A, B$ for the least squares estimators and $a, b$ for their observed values.

   (a) Draw a scatter plot and fit the regression line. See the sample shown.

   (b) What is $a, b$ and the estimated mean strength at $x = .43$ (Why this number?)

   (c) Estimate the variance.

   Use the fitted values $a, b, s^2$ as the fixed parameters of a simulation model. Perform exactly $M = 100$ independent repetitions of the ten-experiment design. Use the random number generator seed as 215 (Why?). Do not resample or change the input levels.

a-solution-plots.png

(d) Compute the empirical means and sample variances of the 100 intercept and slope estimates.

(e) Compare them with the theoretical means and variances under this simulation model.

(f) Plot the simulated slope estimates.

6. **Multiple regression: investigate the computation.** This is a computational exercise; Python packages may be used. Ten observations of four predictors and a response are provided below. Fit the model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + e_i,$$

with fixed inputs and independent $N(0, \sigma^2)$ errors. Include an intercept. Denote the design matrix by $X$ and the observed response vector by $\mathbf{y}$.

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $y$ |
|---|---|---|---|---|
| 0 | 0 | 5 | 7 | 97 |
| 16 | 4 | 1 | 11 | 275 |
| 9 | -3 | 8 | 4 | 246 |
| 4 | 2 | 3 | 9 | 140 |
| 25 | -5 | 10 | 2 | 482 |
| 1 | -1 | 6 | 6 | 122 |
| 9 | 3 | 2 | 10 | 183 |
| 16 | -4 | 9 | 3 | 359 |
| 1 | 1 | 4 | 8 | 82 |
| 4 | -2 | 7 | 5 | 146 |

(a) First attempt to compute the coefficient estimates by solving

$$(X^\mathsf{T} X)\mathbf{b} = X^\mathsf{T}\mathbf{y}$$

using `numpy.linalg.solve`. Record any error or warning. If values are returned, retain them for investigation rather than assuming success.

(b) Obtain an equivalent model with a full-column-rank matrix and fit it. Exhibit two distinct coefficient vectors for the original model that results in the same fitted responses.